

Stoquify Enterprise Sales-to-Cash Readiness

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Scope: Repository audit and target-state design only

Overall status: **BLOCKED**

Confidence: HIGH_FOR_LOCAL_REPOSITORY_TRUTH_LOW_FOR_PRODUCTION

Verdict

The immediate online cash-sale kernel is materially implemented and locally verified, but the enterprise sales-to-cash system is not production-ready.

Strengths

- The authoritative POS commit transaction owns sale completion, payment rows, stock issue, cash-session totals, accounting postings, audit, events, and outbox creation in one database transaction.
- Accounting posting kernels are source-linked, balanced, idempotent, open-period aware, and connected to close invalidation.
- Inventory stock events are organization- and location-scoped, idempotent, version-checked, valuation-aware, and source-linked.
- Refund and void paths preserve history through compensating records.
- Offline ingestion and replay have device, signature, ordering, payload-conflict, idempotency, provisional-receipt, and operator-resolution controls.
- Provider-event ingestion, statement import, suspense, matching, and reconciliation infrastructure exists independently of POS.

Release blockers

1. Electronic POS tenders are marked PAID from operator-entered references without a provider-authoritative authorization, capture, unknown, settlement, reversal, dispute, or chargeback lifecycle.
2. Online POS has no tenant-scoped clientCommitId result registry, so a response-loss retry cannot deterministically return the original committed result.
3. A completed sale does not synchronously bind an immutable receipt/fiscal-source document; fiscal materialization is asynchronous and production worker, authority, and delivery proof are absent.
4. Delivery/on-account order-to-cash lacks authoritative reservation, fulfillment, goods-issue, partial-delivery, invoice, AR, and correction state machines.
5. POS route, navigation, read actions, write actions, offline actions, and seeded roles use inconsistent permission and module-entitlement contracts.
6. Business-day statement, blind tender declaration, manager sign-off, X/Z reporting, statement posting, and accounting-close handoff are not complete authoritative aggregates.
7. PostgreSQL concurrency, provider sandbox/live, fiscal authority, hardware, browser, multi-terminal, and operational recovery evidence is absent or incomplete.

Evidence boundary

Local evidence

- Current dirty worktree source and Prisma schema
- Existing repository audit and evidence artifacts
- Graphify component, action, app, hook, type, and service graphs
- Focused TypeScript, Prisma, Jest, and static/runtime gate results
- Dated official product and security documentation

Not production proof

- No production database, logs, traces, provider credentials, or provider statements were inspected.
- No production receipt signing secret or fiscal authority was exercised.
- No legal or statutory conclusion was made for a country pack.
- No PCI DSS assessment, penetration test, accessibility audit, disaster recovery exercise, or load test was performed.
- No browser/hardware/offline multi-device field test was performed.
- The audited worktree contained user-owned uncommitted changes and therefore is not a release commit certification.

Authoritative owners

Transition	Current owner	Status	Risk/notes
Immediate online POS sale finalization	services/pos/pos.service.ts::commitPOS Sale	IMPLEMENTED_NOT_PRODUCTION_PROVEN	
Offline POS replay into the online finalizer	services/pos/offline-sync.service.ts -> commitPOSSale	IMPLEMENTED_NOT_PRODUCTION_PROVEN	
POS stock issue and valuation	services/inventory/inventory-stock-event.service.ts::postPOSStockIssue	IMPLEMENTED_NOT_PRODUCTION_PROVEN	
Sale, payment, refund, and void ledger posting	services/accounting/postings	IMPLEMENTED_NOT_PRODUCTION_PROVEN	
Fiscal document materialization	services/compliance/fiscalization-outbox.service.ts -> createFiscalDocumentFromPostedSource	PARTIAL_ASYNC_AND_NOT_AUTHORITY_PROVEN	services/pos/pos.service.ts also contains an unused createPOSFiscalDocumentInTx helper, creating ownership ambiguity.
Electronic tender authorization and capture	None	MISSING_FROM_POS	
Delivery/on-account reservation and fulfillment	None	MISSING	

Transition	Current owner	Status	Risk/notes
Business-day retail statement posting	None	MISSING	

Workflow readiness

WF-POS-IMMEDIATE: Immediate online POS cash sale

Status: **PARTIAL**

Implemented:

- Scoped draft cart
- Terminal, session, and drawer validation
- Exact monetary calculation
- Transactional sale completion
- Payment row creation
- Stock issue and valuation
- Sale and payment postings
- Audit, business event, notification, and fiscalization outbox
- Compensating refund and void

Gaps:

- No clientCommitId result replay
- No PostgreSQL race proof
- Receipt/fiscal source remains pending after commit
- Browser, hardware, and live operational proof absent

WF-POS-ELECTRONIC: Immediate POS electronic tender

Status: **RISKY**

Gaps:

- Manual reference validation is treated as successful payment
- POS Payment rows are created as PAID without provider-owned capture truth
- PaymentTransaction and ProviderEvent state are not connected to POS commit
- Unknown, declined, reversed, disputed, chargeback, and settlement states are incomplete

WF-POS-ON-ACCOUNT: Immediate POS on-account sale

Status: **PARTIAL**

Implemented:

- Customer requirement
- Receivable document and customer-ledger creation
- Sale, tax, COGS, and inventory posting

Gaps:

- This is an immediate goods issue with AR, not delivery/on-account order fulfillment
- No promise, reservation, partial shipment, delivery evidence, or shipment-based invoicing

WF-OTC-DELIVERY: Delivery/on-account order-to-cash

Status: **MISSING**

Gaps:

- SalesOrder status names exist but no authoritative confirm, reserve, pick, ship, deliver, or partial-fulfillment commands were found
- No fulfillment aggregate or delivery document
- No reservation consumption contract
- No fulfillment-triggered COGS and inventory posting
- No delivery-triggered invoice and AR orchestration
- No linked return, credit-note, and goods-issue reversal lifecycle

WF-CLOSE: Shift close, business-day statement, reconciliation, and accounting close

Status: **PARTIAL**

Implemented:

- POS session close and variance evidence
- Offline replay/conflict close blockers
- End-of-day readiness read model
- Provider statement import, matching, suspense, and reconciliation
- Accounting close assurance and invalidation

Gaps:

- No authoritative business-day aggregate
- No blind tender declaration aggregate
- Branch payment reconciliation is explicitly unsupported in end-of-day readiness
- Manager sign-off is explicitly unsupported in end-of-day readiness
- No governed X/Z report and statement-posting lifecycle

Capability matrix

ID	Capability	Status	Risk	Evidence
CA P-01	Single authoritative online sale finalizer	IMPLEMENTED	MEDIUM	services/pos/pos.service.ts::commitPOSSale; actions/pos/tender.actions.ts; services/pos/offline-sync.service.ts

ID	Capability	Status	Risk	Evidence
CA P-02	Atomic sale, payment, stock, cash, ledger, audit, and outbox commit	IMPLEMENTED_NOT_PRODUCTION_PROVEN	MEDIUM	services/pos/pos.service.ts; services/pos/ tests /pos.service.test.ts
CA P-03	Tenant-scoped request idempotency and original-result replay	MISSING	CRITICAL	services/pos/pos.schemas.ts has no clientCommitId; commitPOSSale result is not stored in a client-result registry
CA P-04	Provider-authoritative electronic payment lifecycle	RISKY	CRITICAL	POS creates legacy Payment rows as PAID; ProviderEvent and PaymentTransaction infrastructure exists but is not used by POS commit
CA P-05	Provider webhook signature, replay, and redaction controls	IMPLEMENTED_NOT_LIVE_PROVEN	MEDIUM	services/payments/provider-event.service.ts; services/payments/ tests /provider-event.service.test.ts
CA P-06	Provider statement matching, suspense, and reconciliation	IMPLEMENTED_NOT_POS_INTEGRATED	HIGH	services/payments; services/reconciliation; payment:cash-truth:gate passed
CA P-07	Immutable receipt and fiscal-source evidence for every completed sale	PARTIAL	CRITICAL	Fiscalization outbox and FiscalDocument model exist; commitPOSSale returns fiscalDocument null with PENDING watermark; services/pos/pos.service.ts::createPOSFiscalDocumentInTx is unused
CA P-08	Receipt delivery independent from financial completion	IMPLEMENTED_PARTIAL_CHANNELS	MEDIUM	Post-commit delivery failure does not roll back sale; WhatsApp outbox is durable; Print, email, and SMS providers remain placeholders or unavailable
CA P-09	Public receipt token security	LOCAL_READY_RELEASE_BLOCKED	HIGH	Signed, expiring, scoped, revocable tokens; receipt:token:config-gate passed with release enforcement off; Production signing secret not configured in the audited environment
CA P-10	Inventory quantity and COGS at physical goods issue	IMPLEMENTED_FOR_IMMEDIATE_POS	MEDIUM	postPOSStockIssue; inventory:valuation:truth:gate passed; inventory stock-event tests passed
CA P-11	Delivery reservation, partial fulfillment, and goods issue	MISSING	CRITICAL	Generic reserved quantity fields exist; No sales fulfillment aggregate or authoritative workflow found
CA P-12	Balanced, source-linked sale, payment, AR, tax, COGS, inventory, refund, and void postings	IMPLEMENTED_NOT_MAPPING_CERTIFIED	MEDIUM	services/accounting/postings; ledger:close-truth:gate passed; No SYSCOHADA or jurisdiction-specific mapping certification performed
CA P-13	Provider fees, settlement clearing, and chargeback accounting	PARTIAL	HIGH	Payment reconciliation models include fees, suspense, and settlement evidence; POS electronic tender does not create the provider transaction lifecycle
CA P-14	Compensating refund, void, and return evidence	PARTIAL	HIGH	POS full refund and void create linked compensating stock/payment/ledger evidence; Partial refund, delivery return disposition, credit-note, and chargeback lifecycles are incomplete
CA P-15	Offline signed, ordered, replay-safe, conflict-aware events	IMPLEMENTED_NOT_FIELD_PROVEN	HIGH	offline:pos:replay:gate passed 16/16; offline-sync tests passed; No multi-device field or authority proof

ID	Capability	Status	Risk	Evidence
CA P-16	Shift close and counted-cash variance	IMPLEMENTED_NOT_COMPLETE_EOD	HIGH	POS session and drawer close services; shift-close tests passed
CA P-17	Business-day statements, blind declarations, X/Z reports, and sign-off	MISSING_OR_UNSUPPORTED	Critical	End-of-day readiness explicitly reports branch reconciliation and manager sign-off as UNSUPPORTED; No BusinessDay, TenderDeclaration, RetailStatement, XReport, or ZReport aggregate found
CA P-18	Consistent RBAC, module entitlement, fresh authentication, and separation of duties	PARTIAL	Critical	Refund and void require fresh authentication; POS route uses OPERATE_POS while commands use pos.use; Cart and sync actions lack consistent module enforcement; Module inventory reported 408 surfaces, 292 enforcement candidates, and 79 missing-permission classifications
CA P-19	Cashier workflow, offline, hardware, privacy, and accessibility UX	PARTIAL	HIGH	ProfessionalPOSSystem.tsx is 2493 lines; Offline readiness is not connected to checkout; Hardware readiness is synthetic; Mobile cart layout, touch target, focus, privacy, inactivity lock, and handover gaps remain
CA P-20	Delivery/on-account order-to-cash	MISSING	Critical	Sales dashboard is analytical; SalesOrder status enum contains lifecycle labels without command owners; No delivery, fulfillment, partial invoice, or linked correction orchestration found

Target decisions

ADR-SALES-01

Decision: Keep commitPOSSale as the only immediate-sale finalizer and make offline replay call it.

Rationale: Prevents divergent accounting, inventory, receipt, audit, and reconciliation kernels.

ADR-SALES-02

Decision: Introduce a separate delivery/on-account orchestration aggregate that reuses inventory, accounting, receipt, payment, and reconciliation kernels.

Rationale: Delivery sales require promise, reservation, fulfillment, goods issue, invoice, AR, and partial-state semantics that do not belong in immediate POS.

ADR-SALES-03

Decision: Add a tenant-scoped POS client-result registry keyed by organizationId, terminalId, and clientCommitId with a canonical request hash and immutable result envelope.

Rationale: A retry after an unknown response must return the original outcome and reject a conflicting payload.

ADR-SALES-04

Decision: Require provider-authoritative state before an electronic tender can complete a sale; UNKNOWN and PENDING are non-success states.

Rationale: Operator-entered references cannot prove authorization or capture.

ADR-SALES-05

Decision: Bind an immutable fiscal-source snapshot and idempotency key at sale commit, while legal-number assignment and authority submission remain country-pack-governed asynchronous steps.

Rationale: Receipt delivery may fail independently, but completed-sale source truth must never depend on mutable catalog or customer data.

ADR-SALES-06

Decision: Model session close, tender declaration, business-day statement, provider settlement reconciliation, and accounting period close as distinct observable lifecycles.

Rationale: Operational count, retail posting, external cash proof, and period certification have different evidence and authority.

Accounting events

Event	Debits	Credits	Condition
Immediate cash sale	Cash on hand; COGS	Revenue; Tax payable; Inventory	
Electronic capture	Provider clearing	Revenue; Tax payable	Only after provider-authoritative captured/confirmed state
Provider settlement	Bank; Provider fee expense where applicable	Provider clearing	
On-account invoice	Accounts receivable	Revenue; Tax payable	
Physical goods issue	COGS	Inventory	
Customer payment	Cash, bank, or provider clearing	Accounts receivable	
Refund, void, return, fee, dispute, or chargeback	Configured correction accounts	Configured correction accounts	Always source-linked compensating records; never mutation of the original journal

Priority backlog

P0

P0-01: Tenant-scoped POS clientCommitId result replay

Done when:

- Same key and same canonical request hash returns the immutable original result
- Same key with a different request hash is rejected and audited

- Concurrent PostgreSQL attempts produce one sale, one stock issue, one payment consequence set, and one result
- No destructive migration is used

P0-02: Provider-authoritative electronic tender bridge

Depends on: P0-01

Done when:

- POS creates or references PaymentTransaction rather than trusting an operator reference
- Pending and unknown never complete the sale
- Signed webhooks are idempotent and state-monotonic
- Captured, settled, fee, reversal, dispute, chargeback, and suspense evidence tie to ledger and reconciliation

P0-03: Immutable receipt/fiscal-source materialization

Depends on: P0-01

Done when:

- Every completed sale has an immutable source snapshot and hash before the commit result is returned
- Legal numbering remains asynchronous and country-pack-authorized
- Delivery retries cannot alter financial truth
- The unused synchronous fiscal helper is removed or formally adopted so there is one owner

P0-04: POS access-boundary normalization

Done when:

- Navigation, page, read, write, offline, receipt, refund, void, and close surfaces share documented permission and module contracts
- Fresh authentication and maker-checker rules are explicit for material actions
- Cross-tenant and cross-location negative tests pass
- Cashier seed roles are least privilege

P0-05: Release proof for concurrency, secrets, provider, fiscal worker, and offline close

Depends on: P0-01, P0-02, P0-03, P0-04

Done when:

- PostgreSQL race evidence is green
- Release receipt-token secret gate is green
- Provider sandbox/webhook/statement evidence is green
- Fiscal worker retry and dead-letter drills are green
- Offline conflict prevents false close and has a tested operator resolution

P1

P1-01: Delivery/on-account order-to-cash aggregate

Depends on: P0-01, P0-04

P1-02: Reservation, partial fulfillment, goods issue, and reversal contracts

Depends on: P1-01

P1-03: Invoice, AR, partial payment, credit-note, and return orchestration

Depends on: P1-02, P0-03

P1-04: Business day, blind tender declaration, X/Z report, statement posting, and sign-off

Depends on: P0-02, P0-04, P0-05

P1-05: Cashier recovery, inactivity lock, handover, hardware, offline checkout, privacy, and accessibility

Depends on: P0-01, P0-04, P1-04

P1-06: Currency minor-unit, tax-mode, UOM, batch, serial, and availability contracts

Depends on: P1-01

Note: Enable only proven combinations; do not imply unsupported capability.

P2

P2-01: Decompose the oversized POS component and service after contracts are frozen

P2-02: Add performance budgets, queue SLOs, operational dashboards, and recovery runbooks

P2-03: Add non-financial analytics only after transaction-truth events are stable

Deferred

- CRM campaigns
- Loyalty
- Forecasting
- Recommendation engines
- Generic analytics
- Unproven multi-currency
- Unproven inclusive tax
- Unproven POS serial, lot, or UOM selection
- Unproven fiscal certification
- Broad UI beautification

Benchmark sources

System	Reviewed	Reused invariant	Source
SAP S/4HANA	2026-08-17	Sales order, outbound delivery, goods issue, billing, partial fulfillment, and reversal are explicit linked process states.	Official documentation
Microsoft Dynamics 365 Commerce	2026-08-17	Shift close precedes statement calculation/posting; statements create sales, invoices, payment journals, and inventory consequences.	Official documentation
Oracle NetSuite	2026-08-17	Sales orders are non-posting; physical fulfillment posts COGS and inventory; invoices post AR and revenue; immediate assured payment uses cash-sale semantics.	Official documentation
Odoo	2026-08-17	POS session, counted cash, payment method, receipt/invoice, stock movement, correction document, and accounting posting are linked but distinct.	Official documentation

Security references

- [PCI DSS v4.0.1](#) — Scope minimization, no PAN/CVV/PIN in application logs or evidence, secure payment-provider boundary, and qualified assessment requirement.
- [OWASP ASVS 5.0.0](#) — Verification baseline for authorization, authentication freshness, sensitive-data protection, logging, input validation, and API controls.

Verification

Command	Status	Result
<code>git status --short</code>	OBSERVED_DIRTY	User-owned modified and untracked files were present before artifact creation and were preserved.
<code>npm run typecheck</code>	PASSED	Exit 0
<code>npm run prisma:validate</code>	PASSED	Schema valid; exit 0
<code>npx jest --runTestsByPath <8 focused suites> --runInBand</code>	PASSED	8/8 suites and 99/99 tests passed in 40.351 seconds
<code>npm run workflow:assurance:runtime-check</code>	PASSED	7/7 runtime tables, 3/3 migration rows, 0 blockers
<code>npm run inventory:valuation:truth:gate</code>	PASSED_EQUIVALENT_WITH_TEMP_OUTPUT	6/6 checks, 0 blockers
<code>npm run payment:cash-truth:gate</code>	PASSED_EQUIVALENT_WITH_TEMP_OUTPUT	14/14 checks, 0 blockers
<code>npm run ledger:close-truth:gate</code>	PASSED_EQUIVALENT_WITH_TEMP_OUTPUT	10/10 checks, 0 blockers

Command	Status	Result
<code>npm run offline:pos:replay:gate</code>	PASSED_EQUIVALENT_WITH_TEMP_OUTPUT	16/16 checks, 0 blockers
<code>npm run receipt:token:config-gate</code>	PASSED_WITH_WARNING	4/4 checks, 0 blockers, production token secret absent, release enforcement off
<code>npm run module:surface:inventory</code>	REPORT_COMPLETED_WITH_TEMP_OUTPUT	408 surfaces, 367 mapped classifications, 292 enforcement-candidate classifications, 79 missing-permission classifications

Success-criteria assessment

Criterion	Status
Every material sales transition has one authoritative owner and an explicit state contract	NOT_MET
Immediate POS and delivery/on-account sales are separated without duplicating kernels	PARTIAL
Sale retries return the original result safely	NOT_MET
Electronic payments have provider-owned pending, unknown, captured, and settled semantics	NOT_MET
Every completed sale has immutable receipt/fiscal-source evidence independent of delivery	NOT_MET
Inventory quantity and COGS move at the correct physical event	MET_FOR_IMMEDIATE_POS_NOT_MET_FOR_DELIVERY
All material accounting consequences are balanced and source-linked	PARTIAL_LOCAL_EVIDENCE_ONLY
Refunds, voids, and returns preserve original history	PARTIAL
Shift close, statement posting, reconciliation, and accounting close are distinct and observable	PARTIAL
Offline events are signed, ordered, replay-safe, conflict-aware, and resolvable	IMPLEMENTED_NOT_FIELD_PROVEN
Tenant isolation, RBAC, entitlement, fresh auth, redaction, auditability, and separation of duties are tested	PARTIAL
Final report distinguishes local evidence from production proof and lists unresolved blockers	MET_BY_THIS_AUDIT
Phased plan is executable by future Codex runs without guessing	MET_BY_THIS_AUDIT

Reviewer decisions required

Reviewer	Decision
Product	Confirm immediate-sale versus delivery-order scope, partial fulfillment, invoice timing, and supported tender/channel combinations.
Accounting controller	Approve event-to-posting rules, clearing/fee/chargeback accounts, tax timing, AR timing, close gates, and correction policy.

Reviewer	Decision
Inventory controller	Approve reservation, availability, goods issue, return disposition, valuation, negative-stock, lot/serial, and UOM policy.
Payments and treasury	Approve provider state normalization, unknown-payment handling, settlement tolerance, suspense, fees, disputes, and chargebacks.
Security	Approve payment-data scope, provider webhook trust, RBAC/entitlement map, fresh-auth thresholds, maker-checker policy, redaction, and evidence retention.
Country-pack qualified human	Approve dated receipt fields, legal numbering, tax timing, fiscal authority workflow, retention, correction, and offline provisional-document rules.
Operations and support	Approve cashier recovery, offline conflict resolution, shift handover, business-day close, dead-letter ownership, and incident runbooks.
Accessibility	Verify keyboard, focus, screen-reader, touch-target, contrast, responsive cart, error, recovery, and receipt-delivery workflows.
SRE and release	Approve SLOs, alerts, provider/fiscal worker health, backups, restore evidence, load limits, rollout, rollback, and production proof.
AI governance	No AI decision or autonomous financial mutation is required for this sales-to-cash kernel; any future AI assistance must remain evidence-bounded and human-approved.

Certification boundary

This report does not certify production readiness, legal compliance, accounting correctness, PCI DSS compliance, security, or accessibility.